

1. Let $V = \mathcal{C}([0, 1], \mathbb{R})$ be the space continuous functions from $[0, 1]$ to $\mathbb{R}$ equipped with the norm

$$\|f\|_1 = \int_0^1 |f(t)| dt \quad \forall f \in V.$$

Let $L_0 : V \rightarrow \mathbb{R}$ be defined by $L_0(f) := f(0) \quad \forall f \in V$.

- (a) Show that L_0 is a linear mapping.

Let $\alpha, \beta \in \mathbb{R}$ and let $f, g \in V$. We get that

$$L_0(\alpha f + \beta g) = (\alpha f + \beta g)(0) = \alpha f(0) + \beta g(0) = \alpha L_0(f) + \beta L_0(g).$$

- (b) Show that $L_0 \notin V'$.

Having shown that L_0 is linear, what would be missing is the continuity.

If we had $L_0 \in V'$, then there would exist $C > 0$ such

$$|L_0(f)| \leq C \|f\|_1 \quad \forall f \in V \quad (*).$$

Hence, as a consequence, if a sequence $(f_n) \subset V$ is such that $\|f_n\|_1 \rightarrow 0$, then $|L_0(f_n)| \rightarrow 0$. Therefore, to show that $(*)$ does not hold, we will construct a sequence $(f_n) \subset V$ with the property that $\|f_n\|_1 \rightarrow 0$ while $|L_0(f_n)| \not\rightarrow 0$.

In fact, take $f_n(x) := (1-x)^n$ for every $x \in [0, 1]$. Then

$$\|f_n\|_1 = \int_0^1 |f_n(x)| dx = \int_0^1 (1-x)^n dx = \left[-\frac{1}{n+1} (1-x)^{n+1} \right]_0^1 = \frac{1}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

while

$$|L_0(f_n)| = |f_n(0)| = 1.$$

Therefore L_0 is not continuous and hence, $L_0 \notin V'$.

- (c) What does this show for linear mappings involving infinite dimensional normed spaces.
This shows that linear mappings involving infinite dimensional normed spaces are not always continuous.

Try showing that all linear mappings between finite dimensional normed spaces are continuous.

In fact, if $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ is linear, then L is continuous. This trivially follows from the fact that

$$\forall (x, y) \in \mathbb{R}^2 : L((x, y)) = \underbrace{L(x(1, 0) + y(0, 1))}_{\text{linearity of } L} = xL((1, 0)) + yL((0, 1)) = \underbrace{ax + by}_{\text{continuous}}$$

with $a = L((1, 0))$ and $b = L((0, 1))$.

2. Let $\ell^1 := \{x = (x_n) \text{ sequence of real numbers such that } \sum_{n=0}^{\infty} |x_n| < +\infty\}$ be equipped with the norm

$$\|x\|_1 := \sum_{n=0}^{\infty} |x_n| \quad \forall x = (x_n) \in \ell^1.$$

Let $m \in \mathbb{N}$ and set $L_m(x) := \sum_{n=0}^m x_n$ for every $x = (x_n) \in \ell^1$. Show that $L_m \in (\ell^1)'$, the dual space of ℓ^1 and that $\|L_m\|_{(\ell^1)'} = 1$.

Following the previous question, we first need to show that L_m is linear. **This is left to you.**

Now, in order to show that $L \in (\ell^1)'$ and that $\|L_m\|_{(\ell^1)'} = 1$, we will show that:

(i)

$$|L_m(x)| \leq \|x\|_1 \quad \forall x = (x_n) \in \ell^1$$

which will imply that $L_m \in (\ell^1)'$ and $\|L_m\|_{(\ell^1)'} \leq 1$ and then

(ii) $\|L\|_{(\ell^1)'} \geq 1$.

For (i) we get

$$|L_m(x)| = \left| \sum_{n=0}^m x_n \right| \leq \sum_{n=0}^m |x_n| \leq \sum_{n=0}^{\infty} |x_n| = \|x\|_1 \quad \forall x = (x_n) \in \ell^1.$$

For (ii), since $\|L_m\|_{(\ell^1)'} = \sup_{\|x\|_1=1} |L_m(x)|$ it is enough to construct an $x = (x_n) \in \ell^1$ with $\|x\|_1 = 1$ such that $|L_m(x)| = 1$. Set $x = (1, 0, 0, 0, 0, 0, \dots)$. Then $\|x\|_1 = 1$ and $L_m(x) = 1$.

3. Let $V = \mathcal{C}^1([a, b], \mathbb{R})$ be the vector space over $\mathbb{R}$ of functions continuous with their first derivative from $[a, b]$ to $\mathbb{R}$. We consider the mapping $\|\cdot\|_* : V \rightarrow \mathbb{R}$ defined by

$$\|f\|_* := |f(a)| + \int_a^b |f'(t)| dt \quad \forall f \in V.$$

(a) Show that $\|\cdot\|_*$ is a norm on V . **This is left to you!!**

(b) Consider the mapping $T : V \rightarrow V$ by

$$T(f)(x) = \int_a^x f(t) dt \quad \forall x \in [a, b], \quad \forall f \in V.$$

(i) Show that T is well-defined i.e., $T(f) \in V$ for every $f \in V$.

We want to show that the function $T(f)$ is continuously differentiable. Since, f is continuous, it follows from the fundamental theorem of calculus that $T(f)$ is differentiable with

$$(T(f))'(x) = f(x) \quad \forall x \in [a, b].$$

Thus $T(f)'$ is continuous and therefore $T(f) \in V$.

(ii) Show that $T \in \mathcal{L}(V, V)$ with $\|T\|_{\mathcal{L}(V, V)} \leq b - a$.

- T is linear: Let $f, g \in V$ and let $\alpha, \beta \in \mathbb{R}$. Then, for every $x \in [a, b]$ we have that

$$\begin{aligned} T(\alpha f + \beta g)(x) &= \int_a^x [\alpha f'(t) + \beta g'(t)] dt = \alpha \int_a^x f'(t) dt + \beta \int_a^x g'(t) dt \\ &= \alpha \int_a^x f'(t) dt + \beta \int_a^x g'(t) dt = \alpha T(f)(x) + \beta T(g)(x) \end{aligned}$$

Thus, $T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$.

- T is continuous with $\|T\|_{\mathcal{L}(V, V)} \leq b - a$. We want to show that

$$\|T(f)\|_* \leq (b - a) \|f\|_* \quad \forall f \in V.$$

Notice that

$$\|T(f)\|_* = \underbrace{|T(f)(a)|}_{=0} + \int_a^b |T(f)'(x)| dx = \int_a^b |f(x)| dx.$$

Now, we write

$$\begin{aligned} f(x) &= f(a) + \int_a^x f'(t) dt \Rightarrow |f(x)| \leq |f(a)| + \left| \int_a^x f'(t) dt \right| \\ &\leq |f(a)| + \int_a^b |f'(t)| dt \\ \|T(f)\|_* &= \int_a^b |f(x)| dx \leq \int_a^b \left[|f(a)| + \int_a^b |f'(t)| dt \right] dx \\ &= (b - a) \left[|f(a)| + \int_a^b |f'(t)| dt \right] = (b - a) \|f\|_* . \end{aligned}$$

(iii) Show that $\|T\|_{\mathcal{L}(V, V)} = b - a$.

Now, we only need to show that $\|T\|_{\mathcal{L}(V, V)} \geq b - a$. From

$$\|T\|_{\mathcal{L}(V, V)} = \sup_{\|f\|_*=1} \|T(f)\|_* ,$$

it is enough to find $f \in V$ with $\|f\|_* = 1$ such that $\|T(f)\|_* = b - a$. We want take $f \equiv 1$ and hence,

$$\|f\|_* = 1, \quad T(f)(x) = \int_a^x f(t) dt = \int_a^x 1 dt = x - a$$

and

$$\|T(f)\|_* = |T(f)(a)| + \int_a^b |T(f)'(x)| dx = \int_a^b 1 dx = b - a .$$

4. Let

$$V := \left\{ f \in \mathcal{C}([0, 1], \mathbb{R}) : \exists C(f) \geq 0 \mid |f(x)| \leq C(f)x \quad \forall x \in [0, 1] \right\}$$

(a) Show that V is a vector subspace of $\mathcal{C}([0, 1], \mathbb{R})$.

Here we want to show that

$$f + g \in V \quad \text{and} \quad \lambda f \in V \quad \forall f, g \in V, \forall \lambda \in \mathbb{R}.$$

(1)

$$\left. \begin{array}{l} f \in V \Leftrightarrow \exists C(f) > 0 \text{ such that } |f(x)| \leq C(f)x \quad \forall x \in [0, 1], \\ g \in V \Leftrightarrow \exists C(g) > 0 \text{ such that } |g(x)| \leq C(g)x \quad \forall x \in [0, 1]. \end{array} \right\} \Rightarrow$$

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq C(f)x + C(g)x = (C(f) + C(g))x \quad \forall x \in [0, 1].$$

Hence, $f + g \in V$.

(2)

$$\begin{array}{l} f \in V \Leftrightarrow \exists C(f) > 0 \text{ such that } |f(x)| \leq C(f)x \quad \forall x \in [0, 1], \Rightarrow \\ |\lambda f(x)| = |\lambda| |f(x)| \leq |\lambda| C(f)x \quad \forall x \in [0, 1] \Rightarrow \lambda f \in V. \end{array}$$

(b) Prove that the mapping $\|\cdot\|_V : V \rightarrow [0, \infty[$ defined by

$$\|f\|_V := \sup_{0 < x \leq 1} \frac{|f(x)|}{x} \quad \forall f \in V$$

is a norm on V .

This is easy! Only the vanishing property of the norm is tricky in this case. In fact,

$$\|f\|_V = 0 \Leftrightarrow \sup_{0 < x \leq 1} \frac{|f(x)|}{x} = 0 \Leftrightarrow f(x) = 0 \quad \forall x \in (0, 1]$$

What about $f(0)$? Notice that $f(0) = 0$ from the definition of V , by taking $x = 0$. Therefore,

$$\|f\|_V = 0 \Leftrightarrow f \equiv 0.$$

(c) For any $f \in V$ set $L(f)(x) := \begin{cases} \frac{1}{x} \int_0^x f(y) dy & \text{if } x \in (0, 1] \\ 0 & \text{if } x = 0. \end{cases}$

(i) Prove that $L(f) \in V$ for any $f \in V$. (This means that the mapping L is well-defined).

We want to show that $L(f)$ is continuous. Notice that $g : x \rightarrow g(x) := \int_0^x f(y) dy$ is differentiable by the fundamental theorem of calculus (FTC) and hence, is continuous. Also $x \rightarrow \frac{1}{x} \int_0^x f(y) dy = \frac{g(x)}{x}$ is continuous at every $x \in (0, 1]$. Therefore $L(f)$ is continuous at every $x \in (0, 1]$. Now let us show that $L(f)$ is continuous at 0.

This is a bit delicate!

We want to show that $\lim_{x \rightarrow 0} L(f)(x) = L(f)(0) = 0$.

Notice that

$$\lim_{x \rightarrow 0} L(f)(x) = \lim_{x \rightarrow 0} \frac{g(x)}{x} = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x} = \underbrace{g'(0)}_{\text{by (FTC)}} = f(0) = 0.$$

(ii) Prove that $L \in \mathcal{L}(V, V)$ with $\|L\|_{\mathcal{L}(V, V)} \leq \frac{1}{2}$.

You can write $\int_0^x f(y)dy = \int_0^x y \frac{f(y)}{y} dy$.

We first start by showing that L is a linear mapping. So, let $f, g \in V$ and $\alpha, \beta \in \mathbb{R}$.

Let $x \in [0, 1]$.

For $x = 0$ we get $L(\alpha f + \beta g)(0) = 0 = \alpha L(f)(0) + \beta L(g)(0)$.

For $x \in (0, 1]$, we have

$$\begin{aligned} L(\alpha f + \beta g)(x) &= \frac{1}{x} \int_0^x (\alpha f + \beta g)(y) dy = \frac{1}{x} \int_0^x [\alpha f(y) + \beta g(y)] dy \\ &= \alpha \frac{1}{x} \int_0^x f(y) dy + \beta \frac{1}{x} \int_0^x g(y) dy = \alpha L(f)(x) + \beta L(g)(x). \end{aligned}$$

Now, it remains to show that

$$\|L(f)\|_V \leq \frac{1}{2} \|f\|_V \quad \forall f \in V.$$

Let $f \in V$.

$$\|L(f)\|_V = \sup_{0 < x \leq 1} \frac{|L(f)(x)|}{x}.$$

Following the hint, we get that

$$\begin{aligned} |L(f)(x)| &= \left| \frac{1}{x} \int_0^x f(y) dy \right| = \left| \frac{1}{x} \int_0^x y \frac{f(y)}{y} dy \right| \leq \frac{1}{x} \int_0^x y \frac{|f(y)|}{y} dy \\ &\leq \frac{1}{x} \int_0^x y \sup_{0 < t \leq 1} \frac{|f(t)|}{t} dy = \left[\frac{1}{x} \int_0^x y dy \right] \sup_{0 < x \leq 1} \frac{|f(t)|}{t} \\ &= \frac{1}{x} \left[\frac{1}{2} y^2 \right]_0^x \|f\|_V = \frac{x}{2} \|f\|_V. \end{aligned}$$

Therefore,

$$\frac{|L(f)(x)|}{x} \leq \frac{1}{2} \|f\|_V \quad \Rightarrow \quad \|L(f)\|_V = \sup_{0 < x \leq 1} \frac{|L(f)(x)|}{x} \leq \frac{1}{2} \|f\|_V.$$

(iii) Using the function $f(x) = x$, deduce that $\|L\|_{\mathcal{L}(V, V)} = \frac{1}{2}$.

Having proved in (ii) that $\|L\|_{\mathcal{L}(V, V)} \leq 1/2$, it remains only to show that

$\|L\|_{\mathcal{L}(V, V)} \geq 1/2$.

As explained in the lectures, we need to construct a function f such that $\|f\|_V = 1$

and $\|L(f)\|_V = \frac{1}{2}$.

Let us check whether the given function $f(x) = x$ is the right choice.

$$\|f\|_V = \sup_{0 < x \leq 1} \frac{|f(x)|}{x} = \sup_{0 < x \leq 1} \frac{x}{x} = 1 \quad \text{and} \quad L(f)(x) = \frac{1}{x} \int_0^x y dy = \frac{1}{x} \left[\frac{1}{2} y^2 \right]_0^x = \frac{1}{2} x$$

Hence

$$\frac{|L(f)(x)|}{x} = \frac{1}{2} \quad \forall x \in (0, 1] \quad \Rightarrow \quad \|L(f)\|_V = \sup_{0 < x \leq 1} \frac{|L(f)(x)|}{x} = \frac{1}{2}.$$

5. Let $V = \mathcal{C}^1([0, 1], \mathbb{R})$ be the space of continuously differentiable functions from $[0, 1]$ to $\mathbb{R}$.

(a) Say whether the space $(V, \|\cdot\|_\infty)$ is a Banach space. (Hint: consider the sequence of functions (f_n) defined by $f_n(x) := \sqrt{x^2 + \frac{1}{n+1}}$ $\forall x \in \mathbb{R}, \forall n \in \mathbb{N}$.)

Notice that

$$\begin{aligned} f_n(x) - f_m(x) &= \sqrt{x^2 + \frac{1}{n+1}} - \sqrt{x^2 + \frac{1}{m+1}} \\ &= \frac{\left[\sqrt{x^2 + \frac{1}{n+1}} - \sqrt{x^2 + \frac{1}{m+1}} \right] \left[\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}} \right]}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} \\ &= \frac{\frac{1}{n+1} - \frac{1}{m+1}}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} \end{aligned}$$

Hence,

$$\begin{aligned} |f_n(x) - f_m(x)| &= \frac{\left| \frac{1}{n+1} - \frac{1}{m+1} \right|}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} \leq \frac{\frac{1}{n+1} + \frac{1}{m+1}}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} \\ &= \frac{\frac{1}{n+1}}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} + \frac{\frac{1}{m+1}}{\sqrt{x^2 + \frac{1}{n+1}} + \sqrt{x^2 + \frac{1}{m+1}}} \\ &\leq \frac{\frac{1}{n+1}}{\sqrt{\frac{1}{n+1}}} + \frac{\frac{1}{m+1}}{\sqrt{\frac{1}{m+1}}} = \sqrt{\frac{1}{n+1}} + \sqrt{\frac{1}{m+1}} \end{aligned}$$

Thus, $0 \leq \|f_n - f_m\|_\infty \leq \sqrt{\frac{1}{n+1}} + \sqrt{\frac{1}{m+1}} \rightarrow 0$ as $n, m \rightarrow \infty$. Therefore, (f_n) is a Cauchy sequence.

If (f_n) convergence to some f in the norm $\|\cdot\|_\infty$, that is, (f_n) converges to f uniformly, the (f_n) converges pointwise to f . Now notice that, for every $x \in [0, 1]$, we have that $f_n(x) = \sqrt{x^2 + \frac{1}{n+1}} \rightarrow \sqrt{x^2} = |x|$. However, the function $f(x) = |x|$ does not belong to $V = \mathcal{C}^1([0, 1], \mathbb{R})$, because f is not differentiable.

(b) Consider the mapping $N : V \rightarrow \mathbb{R}$ defined by $N(f) := \|f\|_\infty + \|f'\|_\infty \quad \forall f \in V$.

(i) Show that N is a norm on V .

(ii) Show that (V, N) is a Banach space. (Hint: Let (f_n) be a Cauchy sequence in (V, N) . Deduce that (f_n) and (f'_n) are Cauchy sequences in $(\mathcal{C}[0, 1], \|\cdot\|_\infty)$ (which is known to be complete) and hence, converge to f and g respectively in the norm $\|\cdot\|_\infty$. Then, write $f_n(x) = f_n(0) + \int_0^x f'_n(t)dt$ and pass to the limit. Then, deduce that the function f is differentiable with $f'(x) = g(x)$ and conclude that $N(f_n - f) \rightarrow 0$.)

Following the hint we pass to the limit here, $f_n(x) = f_n(0) + \int_0^x f'_n(t)dt$ to get

$$f(x) = f(0) + \int_0^x g(t)dt \quad \forall x \in [0, 1] \quad \Rightarrow \quad f \text{ is differentiable and } f'(x) = g(x).$$

Thus f is continuously differentiable and

$$N(f_n - f) = \|f_n - f\|_\infty + \|f'_n - f'\|_\infty = \|f_n - f\|_\infty + \|f'_n - g\|_\infty \rightarrow 0.$$

6. (a) Let $(V, +, \cdot)$ be a vector space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Define what is meant by the statement that two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on V are equivalent.
- (b) Let $\|\cdot\|_1$ be the norm on $\mathbb{R}^2$ defined by $\|(x, y)\|_1 := |x| + |y| \quad \forall (x, y) \in \mathbb{R}^2$. Let $N : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$N(x, y) := \sup_{t \in [0, 1]} |x + ty| \quad \forall (x, y) \in \mathbb{R}^2.$$

Find two constants $C_1, C_2 > 0$ such that

$$C_1 \|(x, y)\|_1 \leq N(x, y) \leq C_2 \|(x, y)\|_1 \quad \forall (x, y) \in \mathbb{R}^2.$$

To be solved in the next tutorial session next week